Matías Bilkis - PhD

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Short bio

I am a quantum scientist working on **quantum AI**. My research scope is interdisciplinary, touching both physics and computer science. My outreach activity focuses on evaluating **tech as an element of social transformation**.

I hold a **degree in physics** (Universidad Nacional de La Plata), **a PhD in physics** (Grup d'Informació Quàntica - Universitat Autònoma de Barcelona), **co-founder** of the Quantum Machine Learning Group at the Center for Computer Vision (QML-CVC), and director of QutSur (Quantum Technologies del Sur).

Experience

I work as the director of the Quantum Machine Learning research group. My current professional goal is to expand my horizons to build a stable personal economy.

I have **10+ working & research experience** in quantum information theory, quantum computing, quantum machine learning, quantum communications, quantum sensing, NISQ computing, generative models, deep learning, reinforcement learning, quantum algorithms, quantum discrimination, scientific dissemination and programming, continuously-monitored quantum systems, variational algorithms, neural networks, quantum correlations.

My research tools are **strong mathematical & physics knowledge**, creative thinking, team research and **programming.** I am an expert programmer in Python, Mathematica, TensorFlow, Jax, PyTorch and also developing my own software libraries **(check github.com/matibilkis/)**.

I am extremely good at handling big groups of talented people, meeting deadlines, fund-raising, project writing, scientific dissemination, communicating skills and project deliverables.

I am used to provide **talks to big audiences** both at outreach level (e.g. my talk at Sonar festival, or my talk at Biblioteca Camp del Arpa experimentAI cycle) and at academic level (e.g. talks at Max Planck institute, CERN, LIFIA, to mention a few with links). I am also good at coordinating talks with many players: check this round table ("Tecnologías Cuánticas: Oportunidades de cooperación entre el sector privado y el sistema público de CyT en Argentina") with the leads of IBM-Argentina, RIPAISC and UNLP CyT. Check also my podcast interview on my experience in quantum machine learning, creating a research group, encouraging interdisciplinary research and forecasting the social impact of this technology.

I have experience as an activity **organizer** (both in real and virtual worlds). At a research group level, I invite world leaders the **QML-CVC seminar series**, from big companies and institutes around the globe such as Xanadu,

Quantinuum, IBM, Freie University of Berlin, Innsbruck University, Chulalongkorn University, to name a few. Also I have **organized conferences** both in Argentina (CUANTOS 7) and Spain (CVCRD - Computer Vision Conference).

I serve as **referee & reviewer** of scientific articles in Nature machine intelligence, Quantum Journal, IEEE information theory, Physical Review Applied, IEEE quantum week.

Background

- Licenciate thesis **('13-'18)**, *Entropía Condicional y correlaciones cuánticas* and directed by Raúl Rossigonli & Norma Canosa at UNLP. **Graduated with honours**.
- PhD Thesis **('18-'23):** "Decision making in quantum environments", directed by John Calsamiglia at Grup d'Informació Quàntica of Autonomous University of Barcelona. **Graduated with honours**.
- QML-CVC group **('23-now):** it started as a posdoc project, but quickly turned into a whole research group in the Computer Vision Center of Barcelona, a super nice adventure I couldn't have done without Fernando Vilariño.

  This group is currently growing under my co-supervision, and consists of two senior researchers, a postdoc, 2 PhD, 1 master and 3 degree students. We have directed 7+ thesis, produced 5+ research articles and established collaborations with 6+ companies & research institutes.
- QutSur Bridging the gap between South America and quantum technology< ecosystem building and consultancy for large corporations.

Teaching & outreach

- Theoretical and experimental physics I&II ('15-'18) **UNLP physics**
  Introduction to physics, quantum physics II ('18-'22) **UAB Physics**
  Social innovation course ('23-'25) **UAB AI degree**.

- Quantum machine learning in a nutshell course.
  Explaining quantum physics with music course (in process).

- Sonar Festival S+T+ARTS — hackathon winner
  Epica Fura dels Baus - Workshop #19
  #AI and #quantum computing: what is quantum promise
  Coordinator QuantumArts - UAB-Cruilla Chair

Projects & awards

- **CONICET** doctoral scholarship (waived for going to the UAB)
  Aid for predoctoral contracts FPI '17
  Los Alamos - 20K EUR quantum computing '20
  Beca **INVESTIGO** 80K EUR
  Ayuda para contratos pre-doctorales 100K EUR MINECO 2017
- **FCT-23-19711**. *The value of intersections: Ethics, AI, Art and Society.* FECyT. 25,000 Eu (2024-2025)
  FCT-23-19711. GRAIL. *Graph Reasoning for Automatic Interpretation and Learning. Spanish Government.* 150.000Eu (2021-)

Llavors: desarrollo de sensores cuánticos inteligentes (en proceso)
Colaboración Misiones (2M Eur) (GMV-Rovimática): consorcio público privado (concedida)
COFUND Marie Curie- UAB quantum thermodynamics & energy savings.
EURECAT - CVC | industrial PhD (Oriol Balló)

Training of human resources

(check thesis at qml.cvc.uab.es/research)

- PhD students: O. Balló and M. Jarda (ongoing)
- Master students: T. Crosta (10/10 & *mención Masperi AFA, Argentina, best thesis of 2024)* and Y. Cordero (ongoing).
- Undergraduate students: T. Esté (9/10), A. Vargas (8/10), Y. Cordero (10/10), J. Aguilera (9/10), M. Sabater (9/10), M. Delgado (9/10).

Selected research articles

Real-time calibration of coherent-state receivers: Learning by trial and error. M. Bilkis, M. Rosati, R. Morral Yepes, and J. Calsamiglia - Phys. Rev. Research 2, 033295

A semi-agnostic ansatz with variable structure for quantum machine learning - M. Bilkis, M. Cerezo, Guillaume Verdon, Patrick J. Coles, Lukasz Cincio - Quantum Machine Intelligence

Sequential testing for continuously monitored quantum systems - G. Gasbarri, M. Bilkis, and J. Calsamiglia published in Quantum Journal

Challenge-Device-Synthesis: A multidisciplinary approach for the development of social innovation competences for students of Artificial Intelligence - M. Bilkis, J. Kohler, F. Vilariño EDULEARN '24

Hybrid Classical-Quantum architecture for vectorised image classification of hand-written sketches - Y. Cordero, S. Biswas, J. Lladós, F. Vilariño, M. Bilkis

Selected events I participated

IEEE quantum week *panel organizer on quantum applications & transdisciplinary research*, **Experiment with AI: the quantum of AI** - Sant Andreu Library; **QTML '23 CERN** "Machine learning continuously-monitored quantum systems", speaker; **What happens after the doctorate in physics?** Perspectives from teaching, industry and academia - Moderator & organizer; **Como Lake School on complex quantum systems** ('22); Machine Learning quantum many-body physics - Tolouse April '22; **Mestizajes '21** - speaker; **Fermilab** (speaker, '21); **ITW '21 speaker**; **QID 2020(+1)** speaker; **QTML '21** speaker; **Q-TURN-'20** speaker; **ML4Q** Max Planck Erlangen speaker

Personal interests

Sports: Swimming, yoga and running .

Musica guitar (I recently got a boss RC600 loop station, so having a lot of fun exploring it!). I also play the piano and used to sing in the chorus. I've had many bands and musical groups in the past, and love to jam!

Photography: Fujifilm X100F.

Writing: I am currently doing the “curso de narrativa” with Javier Argüello in Escola d’Escriptura **de Barcelona**
Languages: Castellano; English (expert); Catalan (visca el barça!); French (basic), Yiddish (אידדיסש).